INCH-POUND

MIL-S-13949/15C
28 March 1993
SUPERSEDING
MIL-P-13949/15B
12 October 1988

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL AF
(WOVEN ARAMID REINFORCEMENT, EPOXY RESIN, FLAME RESISTANT, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-S-13949

REQUIREMENTS

Reinforcement material A in accordance with MIL-S-13949 Single or multiple layers on double sided metal clad base materials

Resin system F in accordance with MIL-S-13949

Special consideration (third letter designator) G, K, M, N, P, or T in accordance with MIL-S-13949

Type of metal foil Any as specified in accordance with MIL-S-13949

Weight of metal foil H, 1, 2, or 0 in accordance with MIL-S-13949

Solderability Base materials clad with metal foil types A, B, C, D, G, H, J, K, L, and M shall be tested in accordance with MIL-S-13949 Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data

Surface and subsurface imperfections Opaque foreign inclusions less than or equal to 020 inch (0 51 mm) shall not occur more than an average of 4 per square foot or an individual maximum of 16 per square foot Opaque foreign inclusions less than or equal to 040 inch (1 02 mm) and greater than 020 inch (0 51 mm) shall not occur more than 2 per square foot

Bow and twist C or X in accordance with MIL-S-13949

Grade of pits or dents The exterior 1 inch border of both full size sheets and cut panels shall be excluded from the inspection

Extent of qualification. Qualification to third letter designator G or T does not extend to each other

Conditioning of specimens prior to test Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method

DISTRIBUTION STATEMENT A Approved for public release, distribut or is unlimited

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Property values

Dimensional stability (for base material thicknesses less than .020 inch (0.51 mm))

After bake Class A, .00075 inch/inch (0.00075 mm/mm) maximum, class B, .00050 inch/inch (0.00050 mm/mm) maximum

After stress Class A, .00075 inch/inch (0.00075 mm/mm) maximum, class B, .00050 inch/inch (0.00050 mm/mm) maximum

Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm))

Crosswise 40,000 pounds per square inch minimum

Lengthwise 50,000 pounds per square inch minimum

Flammability

Duration of burning time, each flame application

Base material thicknesses greater than or equal to .020 inch (0.51 mm)

Average 25 seconds maximum

Single 30 seconds maximum

Base material thicknesses less than .020 inch (0.51 mm) 15 seconds maximum

Burn length, average 12 inches maximum

CTE, X or Y axis (below the glass transition temperature) 4 to 10 ppm/°C

Peel strength values for copper foil See table I

Arc resistance 60 seconds minimum

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)) 40 kilovolts minimum

TABLE I Peel strength values, minimum pounds per inch of width

Test and copper foil weight	Base material thickness	
	Less than 0.020 inch	0.020 inch and over
After thermal stress		
½-oz/ft² (designator H)	4.0	4.5
1-oz/ft² (designator 1)	5.0	6.0
2-oz/ft² (designator 2)	6.0	7.0
At elevated temperature of 125°C		
½-oz/ft² (designator H)	3.5	4.0
1-oz/ft² (designator 1)	4.0	5.0
2-oz/ft (designator 2)	5.0	6.0
After exposure to processing solutions		
½-oz/ft² (designator H)	3.0	4.0
1-oz/ft (designator 1)	4.0	5.0
2-oz/ft (designator 2)	5.0	6.0

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Electric strength, average, (for thicknesses less than .020 inch (0.51 mm)) 750 volts/mil minimum

Loss tangent at 1 MHz .045 maximum

Permittivity at 1 MHz 4.5 maximum

Surface resistivity

After moisture resistance conditioning 10^4 megohm minimum

After elevated temperature (125°C) conditioning 10^3 megohm minimum

Volume resistivity

After moisture resistance conditioning 10^6 megohm-cm minimum

After elevated temperature (125°C) conditioning 10^3 megohm-cm minimum

Water absorption (for thicknesses greater than or equal to .020 inch (0.51 mm)) 2.0 percent maximum

Supersession information See table II

TABLE II Supersession information 1/

Superseding type designation	Superseded type designation in accordance with MIL-P-13949/15A, 11 February 1988 and MIL-P-13949/15, 19 January 1985
AF*******/*****	AE*******/*****

1/ Asterisks are used in place of identifiers that did not change

Type designation Consists of "AF" prefix followed by coded alphas and numeric characters as shown in the following example

AF N 0310 CH/C1 A 1 C

Base material
Special considerations (G, K, M, N, P or T, see MIL-S-13949)
Nominal base thickness (see MIL-S-13949)
Type of metal foil (side 1, see MIL-S-13949)
Weight of metal foil (side 1, see MIL-S-13949)
Type of metal foil (side 2, see MIL-S-13949)
Weight of metal foil (side 2, see MIL-S-13949)
Grade of pits and dents (A, B, C, or D, see MIL-S-13949)
Class of base material thickness tolerance (1, 2, 3, or 5, see MIL-S 13949)
Class of bow and twist (C or X, see MIL-S-13949)

Revision letters are not used to denote changes due to the extensiveness of the changes

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CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
NSA - S5
DLA - ES

User activities
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5998-0024-04)

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